

2

What Is Technology? Defining or Characterizing Technology

Why Bother with Definitions?

Many students, in my experience, especially in the natural sciences, are impatient with disputes about definitions. They are often called “merely semantic” and may seem hairsplitting. Indeed, they are semantic, in that they deal with meaning, but they are hardly trivial. Many apparently substantive disagreements really stem from the disputants having two different definitions of what is being discussed, say religion, but not being aware of it. Often people think that definitions are purely arbitrary; it means that effort need not be wasted on choosing among opposing or alternative definitions. This is itself based on one view of definition, but it is not the only one. We shall learn something about philosophy by seeing the different sorts of definitions that people have used and their connection to differing philosophical views.

Looking at the alternative definitions of technology shows something about the alternative kinds of definition and also about the characterization of technology. Even if one doesn’t find a final definition on which everyone can agree, an investigation of the definition of technology shows us the range of things that can count as technology and some of the borderline cases where people differ on whether something should be counted as technology or not. Even an unsuccessful search for a best definition helps us to explore the layout of the area we are investigating.

Kinds of Definitions

Let us look at a few different sorts of definitions. At one extreme is the ancient notion of a **real definition**. The ancient Greek philosophers Socrates

WHAT IS TECHNOLOGY?

(470–399 BCE), his student Plato (428–347 BCE), and his student, in turn, Aristotle (384–322 BCE) held to this notion of definition. This view assumes that there is a real structure to the world that corresponds to our words and that a correct definition will match the real nature of things. Socrates went about questioning people about the definition of notions such as justice, courage, or piety and showed the people he questioned how their definitions failed to fit with their notions. Socrates appeared to assume and Plato argued that there is a real nature or structure of justice, of courage, and of piety and that the real definition will fit this. Aristotle claimed that objects have essences and that real definitions will match these. Real definitions of the sort that Plato and Aristotle sought are supposed to “cut nature at the joints”; that is, correspond to the “natural kinds” of things. Some contemporary thinkers view scientific definitions, such as definitions of the chemical elements in terms of atomic weight and number, as true definitions in this sense. Some recent writers in technology studies claim that leading writers on technology in the twentieth century, such as Martin Heidegger (1889–1976) and Jacques Ellul (1912–94), are mistakenly searching for an “essence” of technology. Heidegger, in fact, rejected the traditional account of forms and categories of Plato and Aristotle. Nevertheless, it is true that Heidegger and Ellul do present what they claim is a single, real, core notion of technology.

A different, nearly opposite, view of definitions is that of **stipulative definitions**. This conception is closer to the view of definitions held by many people today. It is claimed that definitions are arbitrary choices or stipulations. Definitions are about words and not things. Opponents of the notion of real definitions deny that there are natural classes or real natures of things to be captured by definitions.

Definitions, on the nominalist view, arbitrarily carve up the world of individuals into classes of things. One can define anything as anything one wishes. Lewis Carroll (1832–1898), who was a logician as well as a writer of children’s books, had Humpty Dumpty hold this view of definitions. Humpty claimed it was a matter of who was master, he or the words. But as some of Humpty’s definitions showed, we cannot sensibly define things in absolutely any way we want. We cannot define religion as a coffee pot and expect to make progress investigating the features of religion. In purely formal systems of abstract math or logic, stipulative definitions make more sense than in common sense or everyday discussions. In an abstract system of math one can lay down a definition and carry it through the system by exact rules of inference. One can use stipulative definitions for the purposes of argument or for a very limited investigation of everyday concepts, but one problem

WHAT IS TECHNOLOGY?

Box 2.1

Nominalism in British philosophy

Nominalists in the late Middle Ages, such as William of Ockham (1285–1347), denied the reality of essences or universals and claimed that only individuals are real. British philosophers of the seventeenth and eighteenth centuries, in the early days of experimental science, held that there were no real definitions. Thomas Hobbes (1588–1679) claimed that definitions are stipulative, even though he thought he could base science on them. In the seventeenth century, Hobbes successfully described the definitional or postulational side of science, but he failed to explain how he tied his definitional and deductive notion of science to observation. For Hobbes, definitions are introduced at the start of an investigation, they are not, as they were for Aristotle, the final result of investigation. John Locke (1632–1704) claimed that we cannot know the real essences of substances. Definitions do not describe essential properties of things or even whether the things defined exist. We can know only nominal essences of substances. In the early eighteenth century David Hume (1711–76) totally denied the existence of real essences, and his position was highly influential upon later empiricism.

with using stipulative definitions in everyday reasoning is that the ordinary meaning of commonly used words sneaks back into the discussion without the author noticing. She slides from her stipulative definition to the ordinary meaning unaware. Writers on technology are, of course, free to define it any way they wish, but they then need to be careful they do not slip back into using other definitions or understandings of technology common in the culture without realizing they have strayed from their original definition.

This leads us to another sort of definition different from both the above, the **reportative definition**. This sort of definition is a report of how people ordinarily use words. It doesn't claim to find the true structure of reality, but it also doesn't simply make up an arbitrary definition by fiat. Dictionary definitions are close to reportative definitions. However, a purely reportative definition would simply describe how people use the word, without legislating "proper" usage. Dictionary definitions contain some normative content. A pure reportative definition could be quite complicated, describing how people in different regions or of different social status use the word. Reportative definitions often have fuzzy boundaries or vagueness of application. Ordinary

WHAT IS TECHNOLOGY?

language is frequently imprecise as to exactly what objects count as falling under the definition. The problem with using reportative definitions of technology is that there are so many different uses of the term around. For instance, some educators associate the word “technology” solely with computers in the classroom, while the school building itself, as well as such older aids to teaching as the blackboard, are part of technology in the broadest sense.

A kind of definition used in philosophy and in other academic areas is a **précising definition**. This sort of definition retains the core ordinary meaning of the word. It is not stipulative or arbitrary. However, unlike a reportative definition it does not simply describe how people actually use the word. It attempts to sharpen up the boundaries of application of the word by describing the range of application and cut-off points. (How big is “big” for a certain kind of thing? How few hairs can one have and still be counted as bald?) Any philosophical attempt at a general definition of technology will be a précising definition.

British empiricist philosophers rejected the existence of essences as real natures of things, but the notion of definition by single defining characteristic continued in general use. In the second half of the twentieth century a number of philosophers concluded that kinds of entities couldn’t be characterized by an essence. One view (of Ludwig Wittgenstein, 1889–1951) is that objects classified under a single name do not share any one single characteristic but share a “family resemblance.” One can often recognize similarities between members of the same human family but cannot find any single feature that they all share. Any pair of things in the class shares some characteristics but no one characteristic is shared by all of them. Wittgenstein gave the example of the notion of “game.” The usual characteristics used to define a game are not shared by every game. Not all games have competing players. Not all games have hard and fast rules, involve equipment or game pieces, and so forth. A game is best defined according to the family resemblance approach by giving paradigmatic examples and suggesting that similar things should also be included. Some contemporary philosophers present views that would make technology an example of a family resemblance concept. Current philosophers of technology such as Don Ihde, Donna Haraway, Andrew Feenberg, and others have ceased searching for an “essence of technology” of the sort propounded by early or mid-twentieth-century thinkers such as Martin Heidegger (see chapter 5) or Jacques Ellul (see below in this chapter, and chapter 7). It is suggested that the things that are included under technology are too varied and diverse to share a single essence.

WHAT IS TECHNOLOGY?

As mentioned above, the major theorists of technology of the first two-thirds of the twentieth century believed that a universal, essential definition of technology could be given. A number of recent theorists, such as Don Ihde, Andrew Feenberg, and others, believe, in contrast, that there is not an essence or single defining characteristic of technology, and that searching for an essential definition is unproductive.

Guidelines for Definitions

Some general guidelines for definition are the following:

- 1 A definition should not be too broad or narrow. (That is, the definition should not include things we would not designate by the word we are defining, and the definition should not be so restricted as to exclude things that should fall under the term defined.)
- 2 A definition should not be circular. (For instance, we shouldn't define "technology" as "anything technological" and then define "technological" as "anything pertaining to technology.")
- 3 A definition should not use figurative language or metaphors.
- 4 A definition should not be solely negative but should be in positive terms. (A purely negative definition in most cases would not sufficiently limit the range of application of the term. A definition by contrast has to assume that the hearer knows the contrasting or opposite term.)

Box 2.2

Philosophical exceptions to the standard rules for definition

The rough guidelines for definitions will have exceptions if one holds certain non-commonsensical philosophical views. For example, some mystics believe God can be characterized only negatively, and hold to so-called "negative theology." Although a simply circular definition is completely unhelpful, it has been pointed out that if one follows out the definitions in a dictionary, looking up the words in the definition, one eventually goes in a circle, although it is a big circle. Some philosophers, such as Hegel, have suggested that the point is not to avoid circularity but to make the circle big enough to encompass everything!

WHAT IS TECHNOLOGY?

An example of defining technology in a too narrow manner is the common contemporary tendency to mean by “technology” solely computers and cell phones, leaving out all of machine technology, let alone other technology. A case of defining technology in a manner that may be too broad is B. F. Skinner’s inclusion of all human activity in technology. Skinner understands human activity as being conditioned and self-conditioning. For Skinner conditioning is considered to be behavioral technology. A related move is the general inclusion of “psychological technology” as part of the motivational apparatus of technological activities, such as chanting in hunter-gatherer societies, or various political beliefs in industrial societies (propagated by propaganda, understood as a kind of technology by Ellul), thereby erasing the distinction between technology and culture by including *all* of culture within technology (see below on Jarvie).

Definitions of Technology

Three definitions or characterizations of technology are: (a) technology as hardware; (b) technology as rules; and (c) technology as system.

Technology as hardware

Probably the most obvious definition of technology is as tools and machines. Generally the imagery used to illustrate a brochure or flier on technology is that of things such as rockets, power plants, computers, and factories. The understanding of technology as tools or machines is concrete and easily graspable. It lies behind much discussion of technology even when not made explicit. (Lewis Mumford (1895–1990) made a distinction between tools and machines in which the user directly manipulates tools, while machines are more independent of the skill of the user.)

One problem for the definition of technology as tools or machines is cases where technology is claimed not to use either tools or machines. One such non-hardware technology is the behavioral technology of the psychologist B. F. Skinner (1904–90). If one considers verbal or interpersonal manipulation or direction of the behavior of another as technology then it appears we have technology without tools. Mumford claims that the earliest “machine” in human history was the organization of large numbers of people for manual labor in moving earth for dams or irrigation projects in the earliest civilizations, such as Egypt, ancient Sumer in Iraq, or ancient China. Mumford calls

WHAT IS TECHNOLOGY?

this mass organized labor “the megamachine” (Mumford, 1966). Jacques Ellul considers patterns of rule-following behavior or “technique” to be the essence of technology. Thus, propaganda and sex manuals will be technology involving rules, and can, but need not always, involve use of tools or hardware.

Technology as rules

Ellul’s “technique” mentioned above is a prime example of another definition of technology. This treats technology as rules rather than tools. “Software” versus “hardware” would be another way to characterize the difference in emphasis. Technology involves patterns of means–end relationships. The psychological technology of Skinner, the tool-less megamachine of Mumford, or the “techniques” of Ellul are not problems for this approach to technology. The sociologist Max Weber (1864–1920), with his emphasis on “rationalization,” resembles Ellul on this, characterizing the rise of the West in terms of rule-governed systems, whether in science, law, or bureaucracy. Physical tools or machinery are not what is central; instead it is the means–end patterns systematically developed.

Technology as system

It is not clear that hardware outside of human context of use and understanding really functions as technology. Here are some examples:

- 1 An airplane (perhaps crashed or abandoned) sitting deserted in the rain forest will not function as technology. It might be treated as a religious object by members of a “cargo cult” in the Pacific. The cargo cults arose when US planes during the Second World War dropped huge amounts of goods on Pacific islands and cults awaited the return of the big “birds.”
- 2 The Shah of Iran during the 1960s attempted to forcibly modernize the country. He used the oil wealth to import high technology such as jet planes and computers, but lacked sufficient numbers of operators and service personnel. It has been claimed that airplanes and mainframe computers sat outside, accumulating sand and dust or rusting, as housing for storage and the operating and repair staffs for them were not made available. The machinery did not *function* as technology.
- 3 Technological hardware not functioning as technology is not solely the province of indigenous societies or developing nations, but can also be

WHAT IS TECHNOLOGY?

present in a milieu of high tech, urban sophisticates. Non-Western technology was displayed in an exhibit of “Primitive [*sic*] and Modern Art” at the Museum of Modern Art as purely aesthetic or artistic phenomena. Indigenous implements and twentieth-century Western abstract art objects were exhibited side by side to emphasize similarity of shape and design. The labels of the primitive implements often did not explain their use, only their place and date. (The use of these devices for cooking, navigation, and other purposes was not explained in the captions.) In some cases neither the museum visitors nor even the curators knew the technological function of the objects. Therefore, although the artifacts were simultaneously both technology and art for their original users, they were not technology, but solely art, for the curators and viewers of the museum exhibit.

These examples suggest that for an artifact or piece of hardware to be technology, it needs to be set in the context of people who use it, maintain it, and repair it. This gives rise to the notion of a **technological system** that includes hardware as well as the human skills and organization that are needed to operate and maintain it (see consensus definition below).

Technology as Applied Science

Much of *contemporary* technology is applied science. However, to *define* technology simply as **applied science** is misleading both historically and systematically. If one understands science in the sense of the combination of controlled experiment with mathematical laws of nature, then science is only some four hundred years old. Even the ancient Greeks who had mathematical descriptions of nature and observation did not have controlled experiment. The medieval Chinese had highly developed technology (see chapter 10) and a rich fund of observation and theory about nature, but had neither the notion of laws of nature nor controlled experiment. Technology in some form or other goes back to the stone tools of the earliest humans millions of years ago. Clearly, with this understanding of science and technology, through most of human history, technology was not applied science. Part of the issue is how broadly one defines science. If one means by science simply trial and error (as some pragmatists and generalizers of Popper’s notion of conjecture and refutation have claimed; Campbell, 1974), then prehistoric technology could be treated as applied science. However, now the notion of science has

WHAT IS TECHNOLOGY?

been tremendously broadened to include virtually all human learning, indeed all animal learning, if one holds a trial and error theory of learning. Perhaps this is an example of a definition of science that is too broad.

Even after the rise of early modern experimental science and the notion of scientific laws in the seventeenth century, and the development of the technology that contributed to the industrial revolution, most technological development did not arise from the direct application of the science of Galileo (1564–1642) and Newton (1642–1727). The inventors of the seventeenth and eighteenth centuries usually did not know the theories of mathematical physics of their day, but were tinkerers and practical people who found solutions to practical problems without using the science of their day. Even as late as Thomas Edison (1847–1931) we find a tremendously productive inventor in the field of electricity who did not know the electromagnetic theory of James Clerk Maxwell (1831–79) and his followers, but who produced far more inventions than those scientists who did know the most advanced electrical field theories. Edison initially even disparaged the need for a physicist as part of his First World War team, thinking one needed a physicist only to do complicated numerical computations, but that a physicist would have nothing much to contribute to technology. By this time Edison's view of the role of theory was getting somewhat dated.

Even in the contemporary situation, in which scientific training is essential for most technological invention, the notion of technology as applied science, if taken in too simple and straightforward a way, is misleading. Modern technology is pursued primarily by those with a scientific background and within the framework of modern science, but many of the specific inventions are products of chance or of trial and error, not a direct application of scientific theory to achieve a pre-assumed goal. Many chemical discoveries have been results of accidents. Safety glass was discovered when a chemical solution was spilled on a piece of glass laboratory apparatus, the glass was accidentally dropped, and it did not break. Penicillin was discovered when a bacterial culture was accidentally contaminated by a mold. Paper chromatography was discovered when a scientist accidentally spilled some chemical on a filter paper, and the chemical separated into two components as it seeped up the paper. The Post-it was discovered when a technologist, Art Fry, using little bookmarks in his hymnal, remembered a temporary glue that a colleague, Spencer Silver, had developed back in 1968 that was too weak to permanently stick two pieces of paper together. In 1977–9 3M began to market the invention, and by 1980 it was sold throughout the USA. Charles Goodyear's development of vulcanization of rubber

WHAT IS TECHNOLOGY?

involved numerous trials and experiments, but one crucial event involved him accidentally leaving his treated “gum elastic” on a hot stove, and noticing that it charred like leather. He then experimented to find a lesser, but optimum, heat of exposure (Goodyear, 1855). Louis Pasteur (1822–95) famously said that chance favors the prepared mind. The development of these accidental discoveries made much use of the scientific knowledge of the people who made them. But the discoveries were hardly the straightforward application of scientific theory to a preset problem.

For these reasons, although technology involves knowledge, particularly know-how, a definition of technology that characterizes it simply as applied science is too narrow.

Systems Definition as a Consensus Definition of Technology

A number of writers have formulated a somewhat complex definition of technology to incorporate the notion of a technological system. The economist John Kenneth Galbraith (1908–2004) defined technology as “the systematic application of scientific or other knowledge to practical tasks” (Galbraith, 1967, chapter 2). Galbraith describes this as incorporating social organizations and value systems. Others have extended this definition to mention the organizational aspect of technology, characterizing technology as “any systematized practical knowledge, based on experimentation and/or scientific theory, which enhances the capacity of society to produce goods and services, and which is embodied in productive skills, organization and machinery” (Gendron, 1977, p. 23), or “the application of scientific or other knowledge to practical tasks by ordered systems that involve people and organizations, living things, and machines” (Pacey, 1983, p. 6). We can combine these definitions into “the application of scientific or other knowledge to practical tasks by ordered systems that involve people and organizations, productive skills, living things, and machines.”

This consensus definition is sometimes characterized as the “**technological systems**” approach to technology. The technological system is the complex of hardware (possibly plants and animals), knowledge, inventors, operators, repair people, consumers, marketers, advertisers, government administrators, and others involved in a technology. The technological systems approach is more comprehensive than either the tools/hardware or the rules/software approach, as it encompasses both (Kline, 1985).

WHAT IS TECHNOLOGY?

The tool approach to technology tends to make technology appear **neutral**. It is neither good nor bad. It can be used, misused, or refused. The hammer can be used to drive a nail or smash a skull. The tool user is outside of the tool (as in the case of carpenters' tools) and controls it. The systems approach to technology makes technology encompass the humans, whether consumers, workers, or others. The individual is not outside the system, but inside the system. When one includes advertising, propaganda, government administration, and all the rest, it is easier to see how the technological system can control the individual, rather than the other way round, as in the case of simple tools.

The notion (known as autonomous technology) that technology is out of human control and has a life of its own (see chapter 7) makes much more sense with technological systems than it does with tools. Technological systems that include advertising, propaganda, and government enforcement can persuade, seduce, or force users to accept them.

As noted above, not all students of technology wish to develop a definition or general characterization of technology. Some, particularly among the “postmodern” devotees of science and technology studies, claim not only that there is no “essence” of technology of the sort that mid-twentieth-century thinkers such as Martin Heidegger, Jacques Ellul and others claimed or sought, but that no general definition of technology is possible.

Despite the validity of the doubts of postmodern students of technology studies concerning an essence of technology, the “consensus definition” delineated above will help to keep the reader roughly focused on the kinds of things under discussion. For instance, the recent advocates of “actor-network theory” (see chapter 12) developed an approach to technology that has many affinities to the consensus definition in the technological systems approach. Advocates of the technological systems approach have recently begun to ally with or even fuse with the social construction of technology approach. Understanding technology as a network fits well with the European sociology of actor-network theory (see box 12.2). Thomas P. Hughes, the person who is perhaps the leading American historian of technological systems, has moved toward the social construction view, and combined it with his own approach (Bijker et al., 1987; Hughes, 2004).

Study questions

- 1 Do you think we can have successful discussions of controversial topics without bothering at all about definitions of major terms?

WHAT IS TECHNOLOGY?

- 2 Can we make words mean anything we wish? In what sense is this true and in what sense is this false?
- 3 Are there any areas of knowledge or subject matters in which there are “real definitions”? Are there any areas in which there are essential definitions?
- 4 What sorts of classes of things might have only “family resemblances” and no essential definitions? (Give examples other than the two given in the chapter and explain your answer.)
- 5 The philosopher Arne Naess (decades later to be the founder of “deep ecology”) in his earliest work surveyed people on the street as to their definitions of various philosophical terms. What sort of definitions was he collecting? Do you think this is a fruitful way to clarify philosophical issues?
- 6 Do you think that the characterization of technology as applied science is correct? Give examples that support this characterization and examples that go against it (other than ones given in the chapter).
- 7 Does the notion of technology without tools make sense? If not, why not? If so, try to give some examples not mentioned in the chapter.